ELEC347 L5 23/10/25 at 9:00

→ Subjects (draft)

→ On Quantisation

→ Bandpass sampling + oversampling

→ FIR
- moving average
- differentiator
- FIR window design

Subjects (draft)

Sampling A/D conversion

- → Aliasing
- → Sampled signal spectrum
- → Anti-aliasing
- → Imaging
- → Anti-imaging (Reconstruction)
- → Quantization SQNR

- → Bandpass sampling
- → Oversampling

FILTERS + Z-TRANSFORM
+ Convolution

FIR - FINITE IMPULSE RESPONSE
- MOVING AVERAGE
- Z-TRANSFORM & CONVOLUTION for filters & signals
- Design - window based
- "optimal"

IIR - INFINITE RESPONSE
- FULL AVERAGE
- Design based on Analogue
  - IMPULSE
  - BZT POLE PLACEMENT

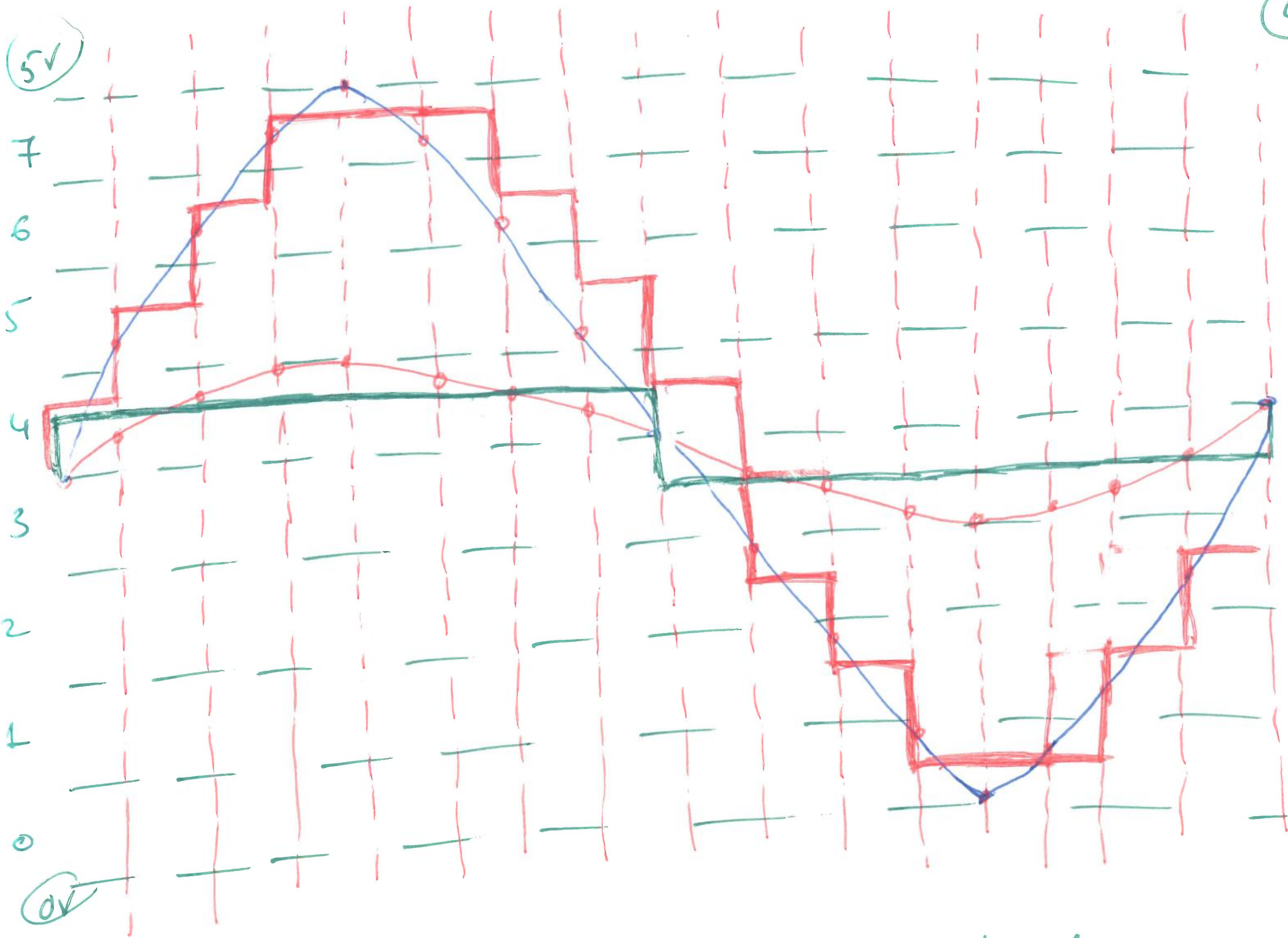
FFT DFT

FAST DISCRETE

BUTTERFLY

PROPERTIES - TIME & FREQUENCY RESOLUTION

FFT OF A REAL SIGNAL



1 bit quantization vs 3 bit quantization

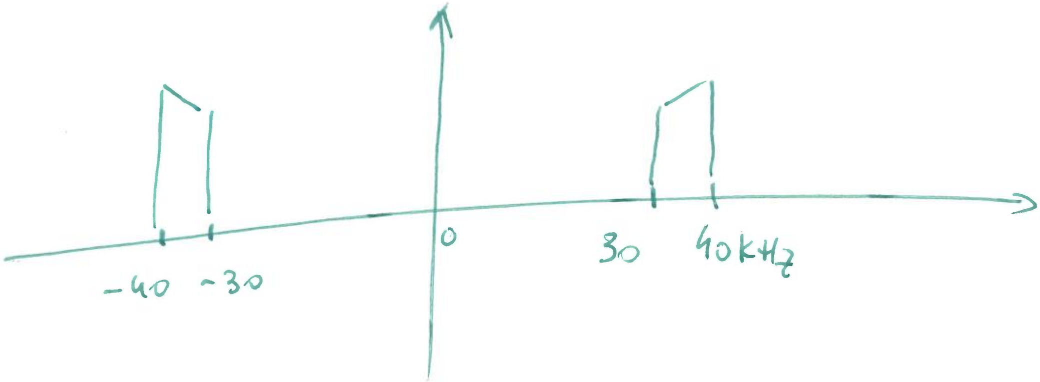
$$QSNR = 6.02B + 1.76\,dB$$

$$= 7.78\,dB, \quad B = 1$$

$$= 6.02 \times 3 + 1.76\,dB$$

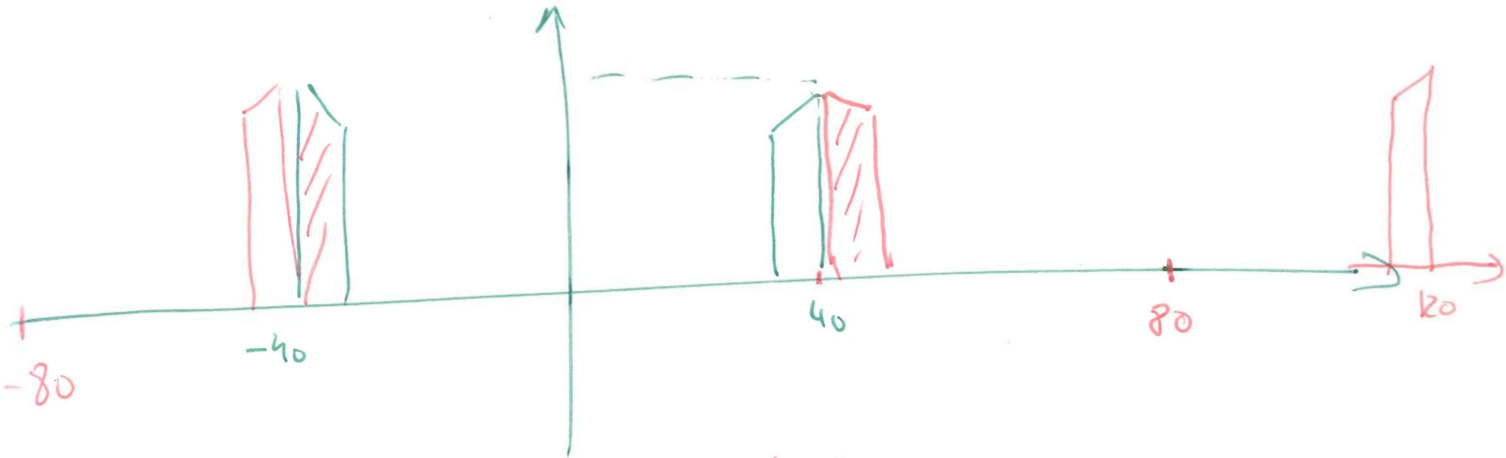
$$= 19.82\,dB, \quad B = 3$$

Bandpass sampling

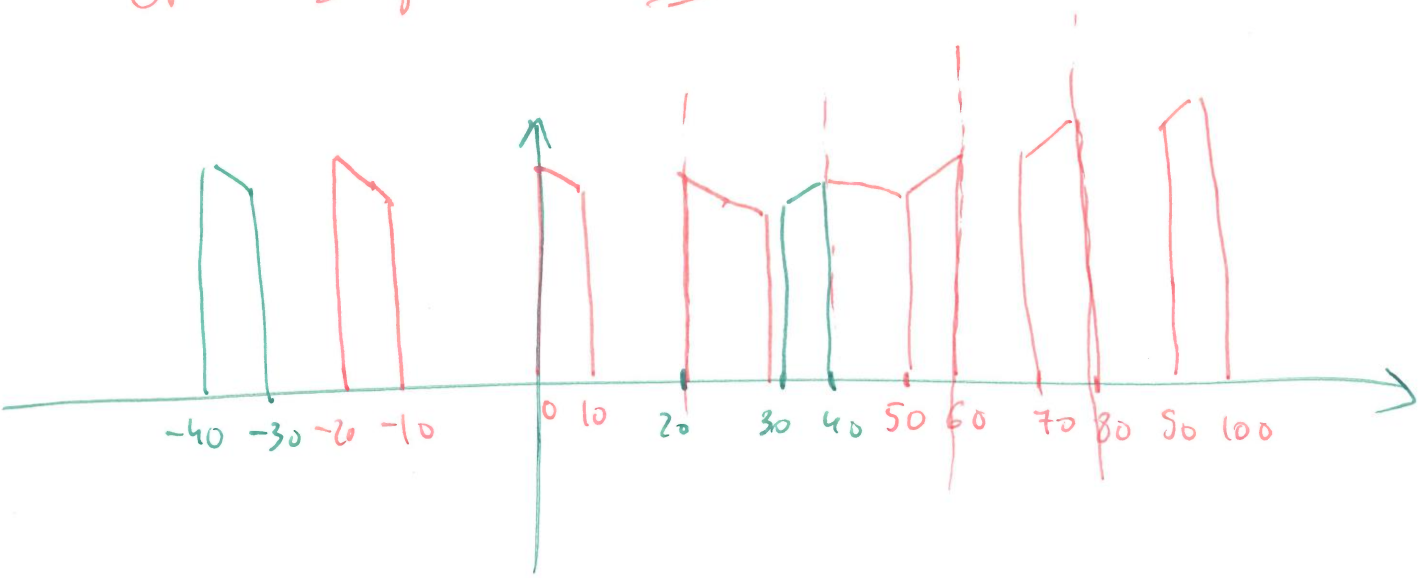


$F_{MAX} = 40$ kHz

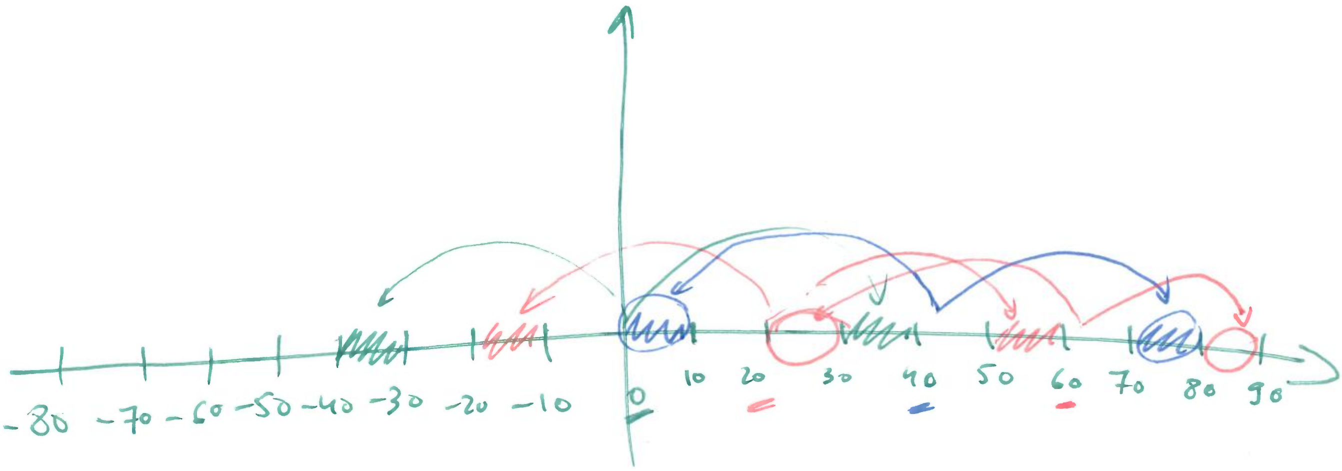
$\Rightarrow \quad F_S \geq 2F_{MAX} = 80$ kHz



or sample at 20 kHz



Bandpass sampling



Interleave without overlapping

$F_S = 20\,kHz$

Sampling

$$\sin 2\pi F t \xrightarrow[t = nT_s]{} \sin(2\pi F n T_s) = x(n)$$

$$y(n) = \frac{x(n) + x(n-1)}{2}$$

$$= \frac{1}{2}\Big(\sin(2\pi F T_s n) + \sin(2\pi F T_s (n-1))\Big)$$

$$= \frac{1}{\cancel{2}}\cancel{2}\sin \frac{2\pi F T_s (n+n-1)}{2} \cos \frac{2\pi F T_s (n-n+1)}{2}$$

$\sin A + \sin B$
$= 2\sin \frac{A+B}{2} \cos \frac{A-B}{2}$

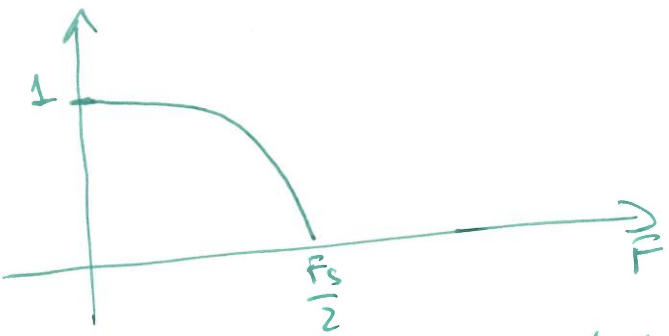
delay 1/2 samples

$$= \sin 2\pi F T_s \left(n - \tfrac{1}{2}\right) \cos \pi F T_s$$

$$F \in \left[0, \frac{F_s}{2}\right]$$

$\cos 0 = 1$

$\cos \pi \frac{F_s}{2} T_s = \cos \frac{\pi}{2} = 0$



Lowpass filter

HW

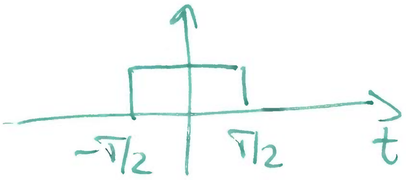
$$y(n) = x(n) - x(n-1)$$

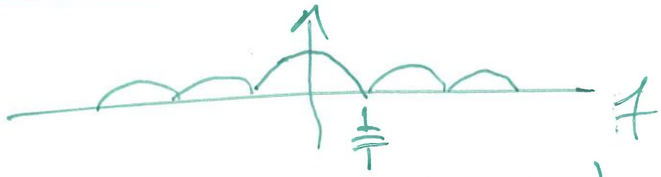
FIR FILTER

$h(0),\ h(1)\ \ldots,\ h(M-1)$ $\qquad$ M coeff

$$y(n) = x(n)\,h(M-1) + x(n-1)\,h(M-2) + \ldots$$
$$+\ x(n-M+1)\,h(0)$$

Design a FIR using the window method





$$\text{rect}\{-\tfrac{T}{2}, \tfrac{T}{2}\} \xleftrightarrow{F} T\,\text{sinc}(\pi f T)$$

$$\text{rect}\{-T, T\} \xleftrightarrow{F} 2T\,\text{sinc}(\pi f\, 2T)$$

$$T/2 \rightarrow T$$

$$\text{rect}\{-F, F\} \xleftrightarrow{F} 2F\,\text{sinc}(\pi\, 2F t)$$

Sample in time $t \rightarrow nT_s = \frac{n}{F_s}$

$$2F\,\text{sinc}(2\pi F n T_s) \xleftrightarrow{F} \sum_k \left[\text{rect}\{-F+kF_s, F+kF_s\}\right]$$

Shift by $\frac{M-1}{2}$ → causal

$$s(n) = 2F\,\text{sinc}\,\frac{2\pi F}{F_s}\left(n - \frac{M-1}{2}\right)$$

$n = 0, \dots, M$
M odd

Hamming window

$$w(n) = 0.5 - 0.5\cos\frac{2\pi n}{M}$$

cos window

up side down

$$h(n) = s(n)\,w(n)$$

$M = ?$ $\quad \frac{B}{F_s} = b \simeq \frac{4}{M}$

3.3 Hamming window

5.5 Blackmann

$F_s = 25\,kHz$

$F = 2{,}5\,kHz$

$B = 1\,kHz$

$$\frac{F}{F_s} = 0.1$$

$$\frac{B}{F_s} = \frac{1}{25} = 0.04 \cong \frac{4}{M}$$

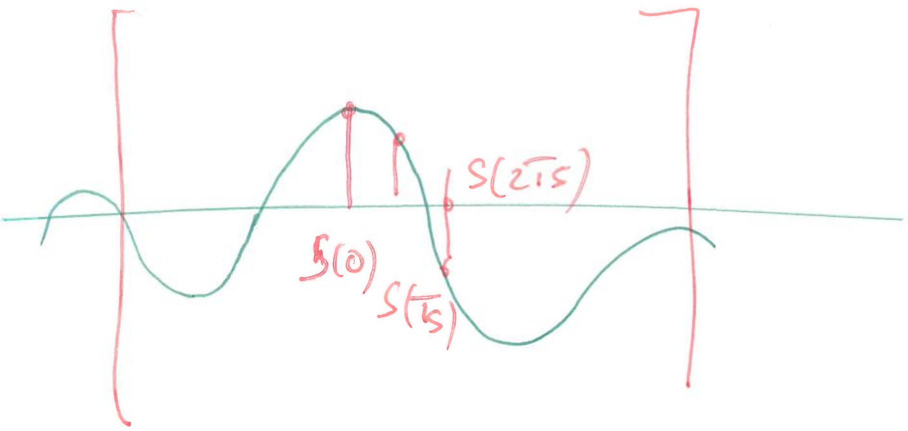
$$\Rightarrow \underline{\underline{M \cong 100}}$$

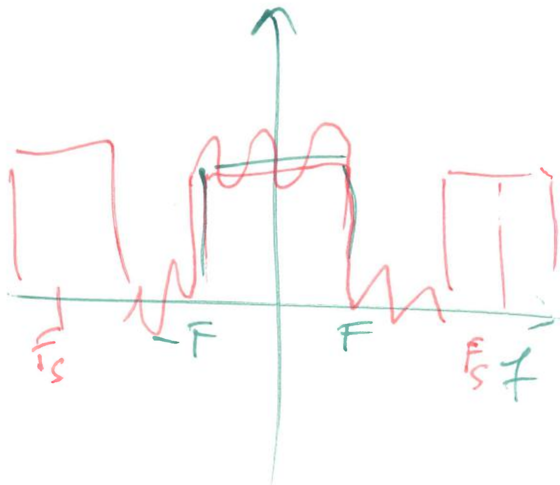
$$h(n) = \operatorname{sinc}\left[\frac{2\pi F}{F_s}\left(n - \frac{M-1}{2}\right)\right]\left(1 - \cos\frac{2\pi n}{M}\right)$$

$$= \operatorname{sinc}\left[0.2\pi\left(n - \frac{M-1}{2}\right)\right]\left(1 - \cos\frac{2\pi n}{M}\right)$$

Normalize

$$h(n) \Big/ \sum_n h(n)$$


S(0)
S(Ts)
S(2Ts)


fs
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